

George Gu

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Education

University of Michigan - Ann Arbor, MI

May 2028

Bachelor of Science in Engineering, Computer Science

GPA: 3.81/4.00

- **Coursework:** Machine Learning, Artificial Intelligence, Reinforcement Learning, Operating Systems, Theory of Computation, Computer Organization, Web Systems, Computer Security, Data Structures & Algorithms, Discrete Math, Linear Algebra

Experience

Capital One

June – August 2026

Software Engineer Intern

Richmond, VA

- Engineered a **production** asset certification feature including a secure **PUT** endpoint under three-tier authorization with a matching **React** status badge and approval flow, eliminating manual certification outreach across **2000+** assets.
- Built **2 EventBridge**-scheduled Lambdas: a daily certification-reminder digest email sender with **SQL** deduplication/time-window logic and a **ServiceNow** CMDB sync uploading reconciliation reports to **S3** and eliminating **8 months** of data staleness from a 1-time load.
- Authored idempotent **Flyway** PostgreSQL migrations adding a CMDB caching layer and tuning SQL for compliance reporting.
- Configured **CI/CD** across dev, QA, and production via Jenkins, IAM roles, SecretsManager and CloudFormation stacks.

BoilerVault Storage LLC

January – June 2026

Founding Engineer (Contract)

West Lafayette, IN

- Built a multi-tenant **FastAPI**/PostgreSQL/Next.js operational platform for a **3-campus** storage business, replacing a failing **Zapier**/Google Sheets stack that desynced bookings from payments, with JWT auth across 20+ REST endpoints and a **318-test** suite.
- Reconciled out-of-order payments via webhook pipelines across **4 Stripe accounts** and **3 WordPress sites**, automating customer, order, and billing creation with **0 manual entry** over **\$130K+** in revenue.
- Automated unmatched charge reconciliation, storing and sweeping oldest-first and saving **3+** hours of manual work per week.
- Migrated 475 legacy bookings into **2,000+** records across 5 tables via three-tier identifier fallback (Stripe ID, email, API lookup).

Nexteer Automotive

May – August 2025

Software Engineer Intern

Saginaw, MI

- Built an **LLM-powered** internal **VS Code** extension in Python and TypeScript parsing **300+** engineering guidelines and generating violation summaries, running **parallel batch** compliance checks across C/H files via Azure AI and Claude APIs.
- Tuned prompt pipelines and few-shot strategies to **95% violation-detection accuracy** on embedded safety-critical steering software.
- Integrated extension into developer workflows, deployed globally, saving engineers up to **8 hours** of manual code review per week.

Villanova University

June – September 2023

Data Engineer

Villanova, PA

- Engineered data pipelines in R, applying unsupervised learning techniques (PCA/H-clustering) to analyze **28,000+** pathogen isolates across **10+ years**, uncovering essential trends published as first author in *Antibiotics*.

Projects

MeloChron | PyTorch, Transformers, tf-musicnn, NumPy, scikit-learn

July – August 2026

- Architected a causal self-attention encoder summarizing track history and 2-layer MLP with **468K parameters** predicting track adoption.
- Provisioned a cloud **RTX 4090** instance to train the **PyTorch** model, cutting a multi-day run to hours while scaling training data **7x**.
- Iterated past a strong baseline (**0.4212**) to an audio-content model at **0.4820 PR-AUC**, evaluated on **100,000** listeners.

Scalable Search Engine | Distributed Systems, Multithreading, Python, Flask, REST, TF-IDF

January – April 2026

- Built a multi-stage MapReduce pipeline in Python to construct an inverted index over **3,000+ documents**, computing TF-IDF scores and document-normalization factors across a parallel, multi-job architecture.
- Served **3 partitioned index segments** from a Flask API, ranking by PageRank-weighted cosine similarity and deployed to AWS.

Resume Screener | scikit-learn, FAISS, NLP, RAG, Streamlit

January – May 2025

- Built an **NLP** pipeline (**TF-IDF** + **LinearSVC**, **scikit-learn**) classifying **12,000+** resumes into 43 categories at **82%/94%** top-1/top-3 accuracy.
- Scored resumes against job postings via **RAG** retrieval (**FAISS** + **LLM**), deployed on **Streamlit** to **100+ users**.

Skills/Additional

Languages: Python, Java, C/C++, SQL, JavaScript, TypeScript, R, HTML/CSS

Technologies: PyTorch, scikit-learn, NumPy/pandas, FastAPI, Flask, Next.js, React, Node.js, REST, PostgreSQL, Stripe, Supabase, Flyway, ServiceNow, Azure, Docker, AWS, Jenkins, Git, Railway, Vercel, FAISS, Streamlit

Concepts: Machine Learning, NLP, RAG, Data Engineering, Backend Development, Distributed Systems, CI/CD, Cloud Computing, Unix/Linux, Shell Scripting, Multithreading, Debugging & Profiling (gdb, valgrind)

Interests: Philosophy and Metaphysics, Game Design, Bouldering, Skateboarding, Golf, Guitar, Calisthenics

Publication

Gu G, Pei H, Zhou A, Fan B, Zhou H, Choi A, Huang Z, "A Comprehensive Study of Historical Detection Data for Pathogen Isolates from U. S. Cattle", 2023, *Antibiotics*, 12, 1509.

doi: <https://doi.org/10.3390/antibiotics12101509>